

SENTHIL KUMAR RADHAMANI

Data Engineer

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SUMMARY

- Experienced data engineering professional with over 20+ years of comprehensive experience in designing, developing, and optimizing data solutions.
- 7+ Years of Experienced Azure Data Engineer with a strong background in designing, implementing and optimizing data solutions on Azure. Proficient in Azure Synapse Analytics, Azure Data Factory (ADF), PySpark, Python. Demonstrated success in developing end-to-end data pipelines, migrating on-premises data infrastructure to the cloud, and leveraging advanced analytics for actionable insights.
- 13+ Years of Experience in performance optimization, data modeling, and ETL/ELT processes using IBM Datastage/Talend with a proven track record of driving business value through data-driven decision-making.

EXPERIENCE

Data Engineer

TD 2024 - Present Toronto, ON

Security Master - Data Integration and Analytics Pipeline

- Define Wealth Data model for Security data from various market data vendors, transform data, generate files, Provide API access to our data and WEB GUI.
- Map and cross-reference security reference data from wealth Sources (BoR Systems) and external sources(e.g. IBM, Bloomberg, Morningstar)
- Built scalable ETL pipelines in Azure Data Factory to ingest and transform data from multiple sources (e.g., SQL databases, APIs, and flat files), processing over 1 TB of data weekly.
- Leveraged Azure Synapse Analytics to design and optimize a data warehouse, improving query performance by 35% through efficient indexing and partitioning strategies.
- Optimized Azure Data Factory pipelines and Azure Synapse SQL queries for performance and scalability, handling a 20% increase in ETL throughput.
- Implemented data quality checks and monitoring within Azure Synapse pipelines, ensuring 99% data accuracy and compliance with organizational standards.

Azure Data Engineer

Royal Bank of Canada(Capital Markets) 2021 - Present Toronto, ON

Global Surveillance System(GSS)

- Developed and implemented data pipelines using Azure Data Factory(ADF) and Azure Synapse resulting in a 30% reduction in processing time.
- Collaborated with cross-functional teams to design and implement scalable data architectures. Migrated Complete GSS(Global Surveillance System) Datastage Project 11.7 to Azure Cloud.
- Migrated data from the on-premises Datastage environment to Azure cloud storage solutions, handling 50 TB of data migration.
- Replicated ETL processes from Datastage to Azure Data Factory pipelines, managing 15 data workflows.
- Integrate Azure Data Factory pipelines with Azure Synapse Analytics to leverage its powerful analytics and data warehousing capabilities resulting in a 30% reduction in job runtime.
- Automated data workflows using Azure Data Factory triggers and schedules, reducing manual processing time by 40% and ensuring timely data availability for downstream analytics.
- Implemented DevOps CI/CD practices for seamless deployment of Azure Data Factory, Azure Synapse Analytics, Python and PySpark solutions ensuring continuous integration.

Data Engineer

Royal Bank of Canada 2017 - 2021 Toronto, ON

Banking Financial Services(BFS)

- Improved data accessibility and performance by leading the design and implementation of data solutions using Azure Data Factory, and PySpark, resulting in a 40% decrease in data processing time.
- Implemented best practices for data modeling, including dimensional modeling and data vault.
- To meet Business needs, User Defined Functions (UDF's) are written.
- Developed JSON Scripts for deploying the pipeline in Azure Data Factory(ADF) that process the data in SQL Activity.
- Managed ETL processes for a large-scale project with 50+ pipelines in Azure Data Factory and Implemented DevOps CI/CD practices for seamless deployment of Azure Data Factory.

STRENGTHS

- Experienced**
Experienced Azure Data Engineer with a strong background in designing and optimizing data solutions.
- Creativity**
Ability to think innovatively and propose novel solutions to challenges in software development.
- Demonstrated Success**
Demonstrated success in developing end-to-end data pipelines and leveraging advanced analytics for insights in Azure Synapse and Azure Data Factory.
- Adaptability**
Quick to learn new technologies and adapt to changing project requirements and environments.

SKILLS

Azure Data Factory(ADF)	Azure Synapse	
Azure SQL DB	Python	PySpark
ETL Datastage	PL/SQL	MarkitEDM

KEY ACHIEVEMENTS

- Recognized for Outstanding Work**
Recognized for outstanding work in developing an efficient data processing system from Cognizant-RBC.
- 30% Reduction in Processing Time**
Azure Data Factory and Azure Synapse resulting in a 30% reduction in processing time.
- Improved Data Accessibility and Performance**
Led the design and implementation of data solutions using Azure Data Factory, and PySpark, resulting in improved data accessibility and performance.

AWARDS

- Best Data Engineering Team**
Received award for exceptional performance and client satisfaction in data engineering project - CDR.
- Employee of the Quarter**
Recognized for outstanding work in developing an efficient data processing system.

ADDITIONAL SKILLS

DevOps (CI/CD)	Talend 7.0	Git/GitHub	
Hadoop/Hive DB	Jenkins	Teradata	T-SQL
JIRA	Tidal Scheduler		

EXPERIENCE

ETL Talend Developer

Scotiabank

2022 - 2023 Toronto, ON

CDOR

- Develop Talend Jobs in Talend 7.0. Migrating jobs from Data stage 11.7/11.5 involves complete lift and shift from Data stage code to Talend.
- Entails processing data from 20 source feeds, performing validations, transformations and loading processed data into HDFS Parquet files.

ETL Developer

Canada Pension Plan Investment Board

2017 - 2017 Toronto, ON

PDS

- Developed mapping and workflows to publish Murex products (CDS, IRS) and MDSA reference data (Equity, bond, party, swaps) into hub.
- Increased DIH publications and subscriptions, resulting in 30% more source files published and delivered to consumers.

ETL Developer

Royal Bank of Canada

2015 - 2016 Toronto, ON

Client Data Repository (CDR)

- Understand data integration needs, and translate business requirements into technical specifications for DataStage ETL solutions.
- Collaborate with business analysts, data architects, and other stakeholders to gather requirements.
- Develop and design Extract, Transform, Load (ETL) processes using IBM DataStage to extract data from multiple sources.

ETL Production Support Analyst

Canadian Imperial Bank of Commerce

2014 - 2015 Toronto, ON

Credit Cards Decision Support Environment(DSE)

- Responsible for the Production Support, maintain Credit Card DSE of Data Stage jobs for CIBC Visa Data warehouse.
- Handling Credit Cards Decision Support Environment (DSE) with 5 team members at offshore.
- Development was performed utilizing Data stage 8.5/7.1, Oracle 10g, PLSQL Stored Procedures, Unix Scripting, SQL Loader, Windows/Unix Sync sort utility and IBM Mainframe access where 95 % of the source files are extracted.

Datstage Developer

Royal Bank of Canada

2010 - 2014 Toronto, ON

BDR (Book Data Repository) /MDR (Market Data Repository)

- Worked as project Lead with 4 team members offshore.
- Participated in complete analysis of existing jobs and developing Data stage 8.5 jobs that involves extracting data from source feeds and perform data transformations/enrichment and deliver to different targets(databases/files).
- Automated complete SOI process to pick up the current market prices from 15 different sources and deliver to IPV systems.
- Co-Ordinated with Business Users to Perform User Acceptance Test (UAT). Prepared Migration and deployment Plans for Production Deployment.

Datstage/PL/SQL Developer

Cognizant/TCS/Infosys

2004 - 2010 Chennai, IN

Senior Associate

- Proficient in writing Oracle PL/SQL that includes functions, stored procedures, triggers and views.
- Improved testing process by implementing unit testing, user acceptance, and support QA testing for 5 software projects.
- Participated in implementing modules in production along with onsite team.

TRAINING / COURSES



Azure Synapse Analytics Essentials:

Completed a comprehensive online course focusing on the core features and functionalities of Azure Synapse Analytics, including data warehousing, big data analytics, and SQL-based querying.



Azure Data Factory

Attended a training program covering the fundamentals of Azure Data Factory, focusing on data integration, ETL process and pipeline orchestration. Gained Hands-on experience in building, deploying and monitoring data workflows.



Python Essentials

Completed a specialized course on PySpark, focusing on its usage for big data processing and analytics tasks, including data parallelism, RDD's, DataFrame operations.

EDUCATION

Bachelor of Engineering in Computer Science

Panimalar Engineering College, Chennai

2000 - 2004 Chennai, India